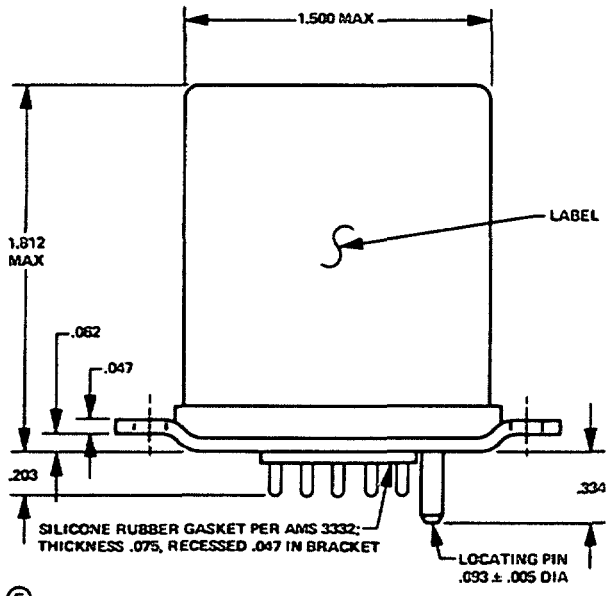
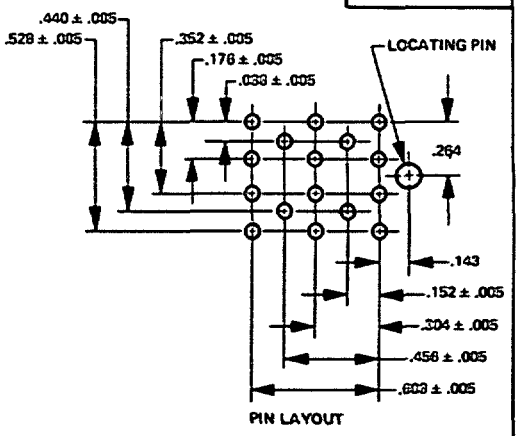
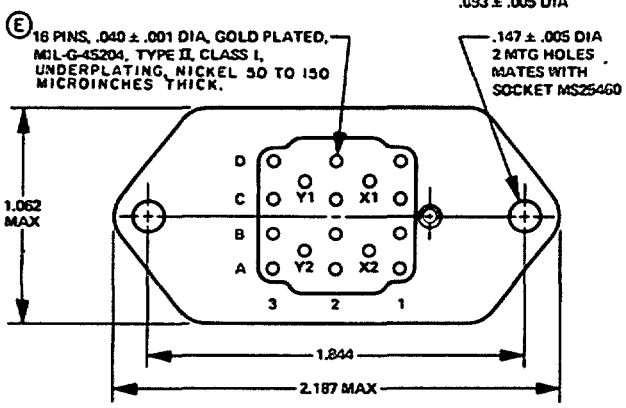
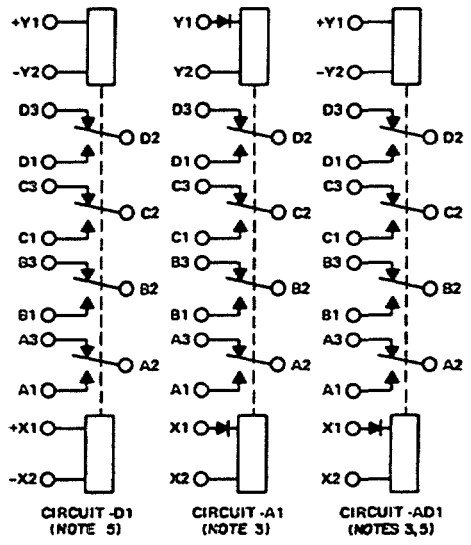
FED. SUP CLASS
5945









NOTES:

1. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .010.
2. TERMINAL NUMBERS NEED NOT APPEAR ON RELAY HEADER PROVIDED THERE IS AFFIXED TO THE RELAY A SUITABLE LEGIBLE CIRCUIT DIAGRAM THAT PERMANENTLY AND POSITIVELY IDENTIFIES EACH TERMINAL LOCATION SPECIFIED HEREON.
3. THE USE OF DIODES ON AC RELAYS IS OPTIONAL. ACTUAL APPLICATION MUST BE SHOWN ON LABEL.
4. PINS TO BE PERPENDICULAR TO HEADER SURFACE WITHIN 1 DEGREE.
5. RELAY IS MAGNETICALLY LATCHED IN BOTH POSITIONS. CAUTION NOTE TO OBSERVE POLARITY MUST APPEAR ON RELAYS WITH DC COILS.
6. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
7. SHOCK, VIBRATION AND ACCELERATION REQUIREMENT APPLICABLE WITH COILS DE-ENERGIZED.

(E) DENOTES CHANGE

MS PART NUMBER	TYPE	COIL	TERMINAL TYPE	MOUNTING OR MATING SOCKET	MAX WEIGHT IN POUNDS
MS25459-D1	I	DC	PLUG IN	MS25460	0.30
MS25459-A1	I	AC	PLUG IN	MS25460	0.32
MS25459-AD1	I	AC-DC	PLUG IN	MS25460	0.32

APPROVED 1 NOV 1960 REVISED (A) 1 MAR 1963 (B) 20 SEP 83 (C) 1 JUL 84 (D) 16 MAR 70 (E) 12 MAY 80

P.A. AIR FORCE 85
Other Cust NAVY - AS

TITLE
RELAY, MAGNETIC LATCH, 5 AMP, 4 PDT, TYPE I (HERMETICALLY SEALED), SOCKET MOUNTED

MILITARY STANDARD
MS 25459

PROCUREMENT SPECIFICATION MIL-R-6106

SUPERSEDES:

SHEET 1 OF 2

DD FORM 1 SEP 83 672

EDITION OF 1 SEP 83 MAY BE USED.

5945-0506

Reviewers: Navy - AS, EC
Air Force 85, 11

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OPERATING CHARACTERISTICS

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MS PART NO. MS25459-	COIL	COIL DATA: Nominal Volts	Nominal Freq Hz	Nominal Res Ω	Max. Volts	Max. Amp	Max Pick-up Voltage: Normal 1/	High Temp Test	Cont Current Test	Drop Out Voltage 1/	TIME-MILLISECONDS - MAX. 6/: Operate	Release	Bounce: Main NO	Main NC	Aux NO	Aux NC
D1	X1,X2 Y1,Y2	28	DC	—	29	0.17	18	18	19.8	N/A	25	N/A	2	2	—	—
A1	X1,X2 Y1,Y2	115	400 4/	—	122	0.07	90	90	95	N/A	25	N/A	2	2	—	—
AD1	X1,X2	115	400 4/	—	122	0.07	90	90	95	N/A	25	N/A	2	2	—	—
	Y1,Y2	28	DC	—	29	0.17	18	18	19.8	N/A	25	N/A	2	2	—	—

RATED CONTACT LOAD (AMPERES PER POLE) CASE GROUNDED

TYPE OF LOAD	LIFE OPERATING CYCLES x 10^3	28 VDC: Main NO	Main NC	Aux NO	Aux NC	115 VAC, 1 PHASE: Main 400Hz	Main 60Hz	Aux 400Hz	Aux 60Hz	115/200 VAC, 3 PHASE 2/: Main 400Hz	Main 60Hz	Aux 400Hz	Aux 60Hz	See Appropriate Notes
RESISTIVE	100	5	5	—	—	5	4	—	—					
INDUCTIVE	100													
INDUCTIVE	20	3	3	—	—	3	2	—	—					
MOTOR	100	1.5	1.5	—	—	1.5	1	—	—					
LAMP	100	0.8	0.8	—	—	0.8	0.6	—	—					
TRANSFER, LOAD														3/
MECHANICAL LIFE REDUCED CURRENT	400	1.25	1.25	—	—	1.25	1	—	—					
MIN CURRENT	APPLICABLE PER SPECIFICATIONS													

ENVIRONMENTAL CHARACTERISTICS

TEMPERATURE RANGE ..... −70°C TO +125°C
MAX ALTITUDE RATING ..... 80,000 FT
SHOCK G - LEVEL ..... 50 G
DURATION ..... 11 mS
MAX DURATION CONTACT OPENING ..... 10 μS
VIBRATION - SINUSOIDAL
G - LEVEL ..... 10 G
FREQUENCY RANGE ..... 5 - 1500 Hz
ACCELERATION ..... 15 G

ELECTRICAL CHARACTERISTICS

INSULATION RESISTANCE, INITIAL ..... 100 MEGOHMS
AFTER LIFE OR ENVIRONMENTAL TESTS ..... 50 MEGOHMS
DIELECTRIC STRENGTH (SEA LEVEL)

	Initial	After Life Tests
COIL TO CASE	1000 VRMS	1000 VRMS
AUX CONTACTS	—	—
ALL OTHER POINTS	1000 VRMS	1000 VRMS

DIELECTRIC STRENGTH (ALTITUDE) 5/

		80,000 FT
COIL TO CASE		500 VRMS
AUX CONTACTS		—
ALL OTHER POINTS		500 VRMS

MAX CONTACT DROP INITIAL ..... 0.150 VOLTS
AFTER LIFE TEST ..... 0.175 VOLTS
OVERLOAD CURRENT ..... 20 AMP
RUPTURE CURRENT ..... 25 AMP
DUTY RATING ..... CONTINUOUS
RFI SPEC MIL-STD-461
(APPLICABLE TO COIL CIRCUITS OF AC OPERATED RELAYS)

NOTES, APPLICATION

1/ OVER TEMPERATURE RANGE
2/ ABSENCE OF VALUE INDICATES RELAY IS NOT RATED FOR 3 PHASE APPLICATIONS
3/ TRANSFER LOAD INDICATES RELAY SUITABLE FOR TRANSFER BETWEEN UNSYNCHRONIZED AC POWER SUPPLIES AT RATING INDICATED.
4/ MS25459-A1 AND AD1 AC COILS MAY BE USED ON 60 Hz IF MAX. AMBIENT TEMPERATURE IS LIMITED TO 85 DEG C, MAX CURRENT WILL BE 0.077 AMP.
5/ WHEN MOUNTED IN MATING SOCKET
6/ WITH NOMINAL COIL VOLTAGE

Reviewers: Navy - AS, EC Air Force 85, 11

APPROVED 1 NOV 1960 REVISED (D) SEE SHTS 1 & 2 (E) FOR CHANGES SEE SHEET 1

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P.A. AIR FORCE 85	TITLE	MILITARY STANDARD
Other Cust NAVY - AS	RELAY, MAGNETIC LATCH, 5 AMP, 4 PDT, TYPE I (HERMETICALLY SEALED), SOCKET MOUNTED	MS 25459
PROCURMENT SPECIFICATION MIL-R-6106	SUPERSEDES:	SHEET 2

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